

Manual J Report



Prepared for Praestholm Residence (Rasmus)

643 Mt Pleasant Ave, West Orange, NJ 07052, USA

Created: Aug 12, 2026

Performed by: njenergyauditor@icloud.com

Calculations approved by ACCA to meet all Manual J 8v2.5 requirements

Heat: 44,067 Btuh **Cool: 25,498 Btuh**

Heating Airflow: **1,343 CFM** Cooling Airflow: **1,040 CFM**

Sensible Cooling: **22,751 Btuh** Latent Cooling: **2,747 Btuh**

Sensible Ratio (SHR): **0.89**

Design Conditions

Weather Station: **ESSEX
COUNTY AP**

#724094, ASHRAE 2021

Latitude: 40.876
Elevation: 173 ft
Elevation Factor: 0.995

⚡ Indoor Heating 70°F

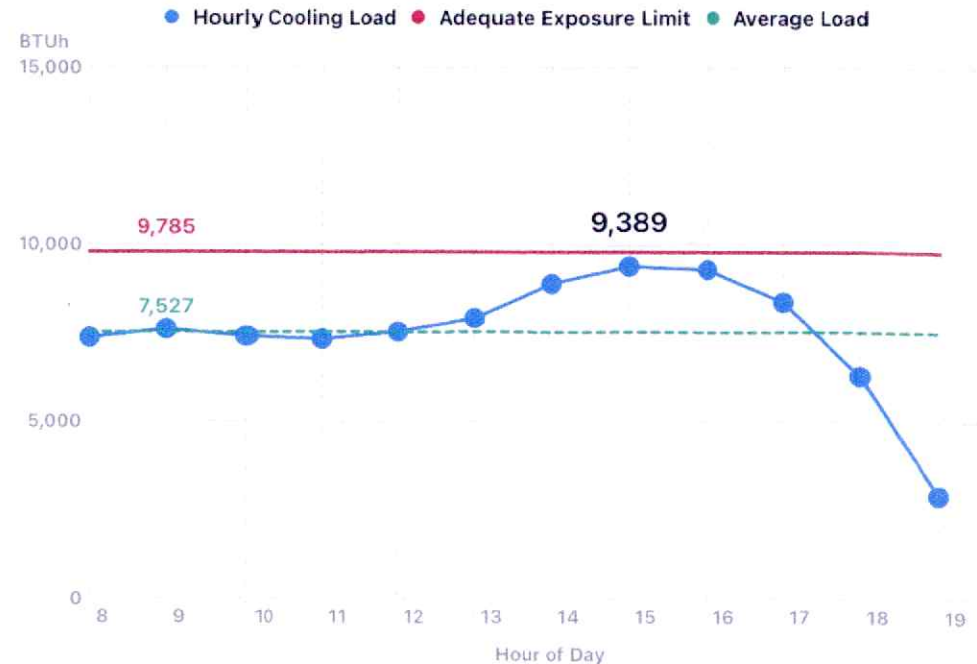
⚡ Indoor Cooling 75°F

⚡ Outdoor Heating 13°F

⚡ Outdoor Cooling 89°F

⚡ Design Grains 27.805

⚡ Daily Range: **Medium**



This application has adequate exposure diversity. The glass areas face a diverse set of directions and/or have sufficient shading (internal or external) to avoid solar gain peaks throughout the day.

Progression 'B'
Scope + Additional

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Tightness 'Loose'

Building Component Summary

Exposed Wall Area: 2,063 sq ft | Conditioned Above Grade Volume: 12,664 cu ft

Component	Details (sq ft)	Heating	Sensible Cooling	Latent Cooling
Floors	978	1,172	97	0
Above Grade Walls*	1,543	9,492	3,958	0
Below Grade Walls*	679	8,804	1,583	0
Ceilings*	1,547	3,946	3,472	0
Windows	152	5,267	5,233	0
Skylights	0	0	0	0
Doors (glass)	43	1,301	1,219	0
Doors (opaque)	39	932	428	0
Infiltration	Tightness: Loose, 2,261 sq ft ACH4 Cooling: 0.44 ACH4 Heating: 0.85	13,154	1,504	1,747
Internal	Occupants: 5 Plants: 0 Appliance Scenario: medium	0	3,550	1,000
Envelope Loads Sub Total		44,067	21,044	2,747
Ventilation	None	0	0	0
Ducts	—	0	0	0
Blower Motor Heat	—	0	1,707	0
AED Excursion	This application has adequate exposure diversity	0	0	0
Grand Total (Btuh)		44,067	22,751	2,747

*Net surface area accounting for doors, windows and skylights

Construction Details (Building Level)

Component	Description Construction ID	Qty	Net Area (sq ft)	Detail	HTM
Floors	Passive flr over tight crawl space/bsmnt(R11Wall) - No floor insul, any finish 19C-0sp	1	264.5	U: 0.368	1.38 0.37
	Bsmnt flr (> 2' below grade) - 32' short slab edge, no insul, any finish 21A-32	3	713.9	U: 0.020	1.13 0.00
Above Grade Walls	2x4 Wood Framed wall, Stucco/Siding - R-0 cavity/R-0 board 12A-0s w	3	247.9	U: 0.240	13.58 6.36
	2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-0 board 12C-0s w	13	762.1	U: 0.091	5.15 1.96
	Garage Partition 2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-0 board 12C-0s w	4	222.0	U: 0.091	5.15 2.89
Below Grade Walls	2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-6 board 12C-6s w	16	310.7	U: 0.060	3.40 0.80
	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	186.8	U: 0.293	16.58 1.68
	Avg depth: 4 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	215.4	U: 0.297	16.82 2.73
	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, Wd Frm R-11, Finished 15B11-0w-x	1	132.9	U: 0.088	4.98 0.68
	Avg depth: 3 ft Garage Partition Bsmnt 8" Stone/Brick Wall - R-0 Board, Wd Frm R-11, Finished 15B11-0w-x	1	143.7	U: 0.086	4.87 1.69
	Ceiling Under FHAvented w/o RB attic - R19 insul 16B-19	7	1,025.3	U: 0.049	2.77 2.44
	Ceiling Under FHAvented w/o RB attic - R21 insul 16B-21	1	271.6	U: 0.044	2.49 2.19
Ceilings	Ceiling Under FHAvented w/o RB attic - R30 insul 16B-30	1	209.9	U: 0.032	1.81 1.59
	Ceiling Under FHAvented w/o RB attic - R56 insul 16B-50	1	40.4	U: 0.020	1.13 1.00
Windows	Northeast Clear Single pane operable window/door 1A-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	3.6	U: 0.900 SHGC: 0.64	50.94 35.72
	Northwest Clear Single pane window w/storm 1C-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.1	U: 0.570 SHGC: 0.56	32.26 28.04
	Northeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	3	19.7	U: 0.550 SHGC: 0.56	31.13 28.40
	Northwest LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.8	U: 0.550 SHGC: 0.56	31.13 28.40
	South LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	20.9	U: 0.550 SHGC: 0.56	31.13 20.72
	Southeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	3	26.6	U: 0.550 SHGC: 0.56	31.13 34.79
	Southwest LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	10.7	U: 0.550 SHGC: 0.56	31.13 34.79
	West LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	18.5	U: 0.550 SHGC: 0.56	31.13 40.55
	Southeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	6.8	U: 0.530 SHGC: 0.56	82.49 68.99
	Southwest LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	26.8	U: 0.530 SHGC: 0.56	34.50 39.67
Skylights	-	0	0	-	0 0
Doors (glass)	Northeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	43.4	U: 0.530 SHGC: 0.56	30.00 28.10
Doors (opaque)	Wood - Solid core w/metal storm 11F	1	21.5	U: 0.280	15.85 7.28
	Metal - Fiberglass core 11J	1	17.4	U: 0.600	33.96 15.60

● Heating HTM ● Cooling HTM

Below Grade

Basement Storage Side

Heat: 2,941 Btuh
Cool: 822 Btuh

Latent Cooling	113
Sensible Cooling	710
Sensible Ratio (SHR)	0.86
CFM Heating	90
CFM Cooling	32

Basement Utility Side

Heat: 10,163 Btuh
Cool: 1,970 Btuh

Latent Cooling	330
Sensible Cooling	1,640
Sensible Ratio (SHR)	0.83
CFM Heating	310
CFM Cooling	75

Basement Bathroom

Heat: 605 Btuh
Cool: 187 Btuh

Latent Cooling	17
Sensible Cooling	170
Sensible Ratio (SHR)	0.91
CFM Heating	18
CFM Cooling	8

1st Story

Exterior Front Picture

Heat: 4 Btuh
Cool: 3 Btuh

Latent Cooling	0
Sensible Cooling	3
Sensible Ratio (SHR)	1.00
CFM Heating	0
CFM Cooling	0

1F Bathroom

Heat: 645 Btuh
Cool: 349 Btuh

Latent Cooling	30
Sensible Cooling	319
Sensible Ratio (SHR)	0.91
CFM Heating	20
CFM Cooling	15

Front Corner Bedroom

Heat: 7,241 Btuh
Cool: 3,664 Btuh

Latent Cooling	325
Sensible Cooling	3,339
Sensible Ratio (SHR)	0.91
CFM Heating	221
CFM Cooling	153

Living Room & Foyer / Hallway

Heat: 3,977 Btuh
Cool: 2,619 Btuh

Latent Cooling	118
Sensible Cooling	2,501
Sensible Ratio (SHR)	0.96
CFM Heating	121
CFM Cooling	114

Rear Corner Bedroom

Heat: 2,977 Btuh
Cool: 1,696 Btuh

Latent Cooling	144
Sensible Cooling	1,552
Sensible Ratio (SHR)	0.92
CFM Heating	91
CFM Cooling	71

Kitchen & Dining

Heat: 7,518 Btuh
Cool: 4,216 Btuh

Latent Cooling	210
Sensible Cooling	4,006
Sensible Ratio (SHR)	0.95
CFM Heating	229
CFM Cooling	183

2nd Story

2/FA Cape Stairwell

Heat: 275 Btuh

Cool: 96 Btuh

Latent Cooling	20
Sensible Cooling	76
Sensible Ratio (SHR)	0.79
CFM Heating	8
CFM Cooling	3

Cape Right Side Bedroom

Heat: 2,881 Btuh

Cool: 1,253 Btuh

Latent Cooling	178
Sensible Cooling	1,075
Sensible Ratio (SHR)	0.86
CFM Heating	88
CFM Cooling	49

Cape Bathroom

Heat: 470 Btuh

Cool: 220 Btuh

Latent Cooling	26
Sensible Cooling	193
Sensible Ratio (SHR)	0.88
CFM Heating	14
CFM Cooling	9

Office, Etc

Heat: 4,371 Btuh

Cool: 2,147 Btuh

Latent Cooling	236
Sensible Cooling	1,911
Sensible Ratio (SHR)	0.89
CFM Heating	133
CFM Cooling	87

General

 Stories 2

 Bedrooms 4

 Estimated Living Area 1,604 sq ft

 Year Built 1946

 Attached Garage: **Open**
minutes per day

 Natural gas? 

Existing heat pumps? 

 Conditioned Basement? 

Room over garage? 

 HVAC System Type: **Duc...**

Building Type: **Single Family**

Internal

 Occupants 5

Scenario: **2400 btuh - Standard Kitchen, utility and lighting**

Heat: 0 Btuh
Cool: 4,550 Btuh

Latent Cooling 1,000
Sensible Cooling 3,550
Sensible Ratio (SHR) 0.78

Plants 12" or shorter 0

Plants 12" - 24" 0

Plants 24" or taller 0

Infiltration

Method: **Tightness**

 Envelope Tightness: **Loose**

Fireplaces 1

 Infiltration Area 2,261 sq ft

Building Type: **Single Family**

Heat: 13,154 Btuh
Cool: 3,250 Btuh

Latent Cooling 1,747
Sensible Cooling 1,504
Sensible Ratio (SHR) 0.46

Would measure ≥ 10 ACH50 via blower door test. Common for homes built pre-1950.

Scanned Living Area: 2,261 sq ft
Estimated Living Area: 1,604 sq ft

Ventilation

Type: **None**

Heat: 0 Btuh
Cool: 0 Btuh

Latent Cooling 0
Sensible Cooling 0
Sensible Ratio (SHR) 0.00